CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (IT) Examination

May 2014

IT403 Data Compression

Date: 20.05.2014, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Define the following Terms (Any Five). **[05]**

1. Entropy
2. Self Information
3. Modeling
4. Uniquely Decodable Property
5. Coding
6. Compression Ratio

Q – 2 Answer following questions. (Any Three) **[15]**

(a) How to measure the performance of multiple Data Compression algorithms? Explain parameters to select one algorithm out of many. 5

(b) An alphabet S ={ a_1, a_2, a_3, a_4, a_5} symbols with probabilities as $P(a_1)= 0.4$, $P(a_2)=0.3$, $P(a_3)=0.2$, $P(a_4)=0.09$, and $P(a_5)=0.01$, Find out Huffman code, source entropy, average length and compressoin ratio. 5

(c) An alphabet S ={ a_1, a_2, a_3, a_4, a_5} symbols with probabilities as $P(a_1)= 0.25$, $P(a_2)=0.25$, $P(a_3)=0.2$, $P(a_4)=0.15$ and $P(a_5)=0.1$. Find out Ternary Huffman code for the above probabilty distribution. 5

(d) Explain Different types of models in data compression. 5

Q – 3 Answer following questions.(Any Three) **[15]**

(a) Explain the encoding process of Adaptive Huffman Algorithm. 5

(b) Differentiate following: 1. Lossy Compression vs. Lossless Compression 5
2. Statistical vs. Dictionary based compression

(c) Consider a source containing 26 distinct symbols [A-Z]. Encode given sequence of symbols using Adaptive Huffman algorithm. 5

Symbol Sequence: MUMMY

(d) Given source of symbols with probabilities as P(A)=0.45 P(B)=0.25, P(C)=0.15, P(D)=0.15. Perform encoding of "BCADB" using arithmetic coding and generate tag. 5

SECTION – II

Q – 4 Find out the correct Match. [05]

A	B
1. Spatial to Frequency Domain Conversion	A. Quantization Table
2. Video Compression	B. Zigzag Scanning
3. Image Quality Control	C. Discrete Cosine Transform
4. Color Space Transform	D. MPEG
5. Lossless Encoding	E. Run Length Coding Algo.
	F. RGB-> YUV

Q – 5 Answer following questions. (Any Three) [15]

(a) Encode following sequence using LZW Coding technique. 5

Sequence: RATATATATRARATRAATATA

(b) Explain Digram Coding compression with suitable example. 5

(c) Encode following string using of LZ78 algorithm. 5

String: Luke Luck takes licks in lakes<END>

(d) Compare & contrast: 5

1. LZ78 and LZW Algorithms.

2. Static Dictionary Based Algorithm vs. Dynamic Dictionary Based Algorithm

Q – 6 Answer following questions. [15]

(a) Find the storage size of Gray scale video clip of 20 second duration with 640x480 resolution @ 30 FPS. 3

(b) **Write short note on following:(Any Three)** 12

1. Sliding Window Compression Techniques
2. Sampling and Quantization of an Audio Signal
3. Spatial and Temporal Redundancies in MPEG Compression
4. Baseline JPEG Compression.

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